So $5\frac{1}{2} - 3\frac{1}{4} = (5-3) + \left(\frac{1}{2} - \frac{1}{4}\right)$

$= 2 + \left(\frac{2}{4} - \frac{1}{4}\right)$ (Since the equivalent fraction of $\frac{1}{2}$ is $\frac{2}{4}$)

$= 2 + \frac{1}{4} = 2\frac{1}{4}$ litres

Example 15 Simplify: $9\frac{1}{4} - 3\frac{5}{6}$

Solution Here 9 > 3 and $\frac{1}{4} < \frac{5}{6}$, So we proceed as follows:

We convert the mixed fraction into improper fraction and then subtract.

$9\frac{1}{4} = \frac{(9\times4)+1}{4} = \frac{37}{4}$

and $3\frac{5}{6} = \frac{(3\times6)+5}{6} = \frac{23}{6}$

Common multiple of 4 and 6 is 12.

Now, $\frac{37}{4} - \frac{23}{6} = \frac{37\times3}{12} - \frac{23\times2}{12}$

$= \frac{111}{12} - \frac{46}{12} = \frac{65}{12} = 5\frac{5}{12}$

Try these

i) Find the sum of $5\frac{4}{9}$ and $3\frac{1}{6}$

ii) Subtract $7\frac{1}{6}$ from $12\frac{3}{8}$.

iii) Subtract the sum of $6\frac{1}{6}$ and $3\frac{1}{5}$ from the sum of $9\frac{2}{3}$ and $2\frac{1}{2}$.

1.6 Multiplication of Fractions

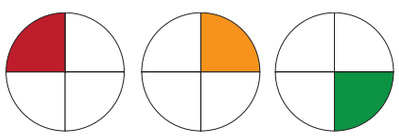
Think about the situation 1 (Multiplication of a fraction by a whole number)

Sunitha wanted to give $\frac{1}{4}$ kg of sweets to each of her 3 friends. So she went to a sweet stall and she asked the salesman to give three $\frac{1}{4}$ kg packets of sweets, how much sweet did she buy?

Solution Weight of three $\frac{1}{4}$ kg packets of sweets $= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{1+1+1}{4} = \frac{3}{4}$ kg



We can illustrate this in the following diagram. In each of these the shaded part is $\frac{1}{4}$ of a circle. Can you find the shaded parts in 3 circles together?



To know the fraction of shaded part in all the three circles, we add the fractions which are represented in each of the circles. So $3\times\frac{1}{4} = \frac{3}{4}$.

The adjacent circle represents $\frac{3}{4}$ parts of the circle.



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